
Connectivity-Adaptive Ansatz Growth for Variational Quantum Dynamics Simulations on Fixed Qubit Topologies

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(code, logs, and per-gate compliance audits accompany this manuscript)

Abstract

Adaptive variational quantum dynamics simulation (AVQDS) tracks real-time quantum dynamics with circuits that are orders of magnitude shallower than Trotterization, but it selects multi-qubit rotations between arbitrary qubit pairs, while near-term quantum processors couple only sparse, fixed pairs of qubits. We introduce connectivity-adaptive AVQDS (CA-AVQDS), which makes the *ansatz-growth step itself* aware of a fixed qubit topology: candidate operators are scored by the same McLachlan-distance criterion as AVQDS, but among candidates whose distance reduction is within a guaranteed fraction β of the best available, the operator with the lowest physical CNOT cost under the current qubit embedding is appended; unavoidable non-local rotations are routed with persistent SWAPs whose placement minimizes the future routing cost of the operator pool. The variational accuracy control of AVQDS—ansatz expansion until the McLachlan distance falls below a threshold—is preserved unchanged, and switching the selection layer off recovers AVQDS exactly. On quenches of the mixed-field Ising model and ramps of the Lieb–Schultz–Mattis chain, mapped to a 2×4 square lattice and to an 8-qubit heavy-hex subgraph that admits no Hamiltonian path, CA-AVQDS produces fully topology-compliant circuits (audited per gate) at final-time fidelities within 0.06% of unconstrained AVQDS, while using 40% fewer CNOTs on the heavy-hex subgraph than a strong baseline that routes the unconstrained circuit with an exhaustively optimized embedding. On topologies whose connectivity covers the Hamiltonian interaction graph, cost-regularized growth yields circuits with $0.52\text{--}0.65\times$ the CNOT count of unconstrained AVQDS itself at matched accuracy—revealing that the standard McLachlan growth criterion leaves a large gate-count degeneracy that connectivity-aware selection can harvest. We provide mechanism, failure, and computational-cost analyses, and demonstrate transfer to a third (linear-chain) topology.

1 Introduction

Simulating the real-time dynamics of interacting quantum systems—preparing $|\psi(t)\rangle = e^{-i\hat{H}t}|\psi(0)\rangle$ and measuring its observables—is among the earliest proposed and most natural applications of quantum computers [1, 2, 3, 4]. Product-formula (Trotter) circuits implement this evolution with rigorous error bounds [5, 6, 7], but their depth grows linearly in simulated time, far beyond what noisy intermediate-scale quantum (NISQ) hardware sustains [8]. Variational quantum dynamics simulation (VQDS) methods instead evolve the parameters of a compact ansatz according to a time-dependent variational principle [9, 10, 11, 12], and *adaptive* variants grow the ansatz on the fly. Adaptive variational quantum dynamics simulation (AVQDS) [13] appends Pauli rotations chosen to minimize the McLachlan distance whenever that distance exceeds a threshold, producing circuits with up to two orders of magnitude fewer two-qubit gates than first-order Trotterization at matched fidelity.

These gate counts, however, are quoted for a fictitious device: AVQDS, like its static counterpart ADAPT-VQE [14, 15], assumes all-to-all connectivity, in which any two-qubit Pauli rotation costs two CNOTs [16, 17]. Real superconducting platforms couple only fixed, sparse pairs of qubits—square-like lattices on Sycamore-class processors [18] and degree- ≤ 3 heavy-hex lattices on IBM’s fleet [19, 20], the same lattices on which utility-scale simulations of kicked-Ising dynamics were recently performed [21]. Two-qubit gates between uncoupled qubits must be synthesized through SWAP networks inserted by a routing pass [22, 23, 24, 25], each SWAP costing three CNOTs. For structured circuits, routing overhead routinely multiplies two-qubit gate counts by $2\text{--}3\times$ on sparse topologies [26, 27]; for the adaptively grown circuits of AVQDS we find the same factor (up to $2.54\times$, Sec. 5). Because the entire premise of adaptive variational dynamics is to spend as few two-qubit gates as possible, an overhead of this size erodes precisely the advantage the method exists to deliver.

The mismatch is not fundamental. The AVQDS growth step must repeatedly choose one operator from a pool, and the authors of AVQDS already observed that the McLachlan-distance scores of competing candidates are frequently near-degenerate [13]. Degeneracy is wasted freedom: an oblivious argmax spends it on whatever operator happens to sort first, while a connectivity-aware rule can spend it on operators that are cheap on the device at hand. This observation motivates our method, *connectivity-adaptive AVQDS* (CA-AVQDS), which moves topology awareness inside the ansatz-growth loop rather than leaving it to a post-hoc compiler. CA-AVQDS combines three contained components layered on an unmodified AVQDS core:

1. **Pool-structure embedding.** An initial spin-to-qubit mapping is chosen (exhaustively, at this scale) to maximize the number of pool operators that act on topology edges and, secondarily, to minimize the total routed excess of the remainder.
2. **Cost-regularized operator selection.** Growth candidates are evaluated with AVQDS’s exact trial-expansion mathematics. Candidates that would immediately bring the McLachlan distance below its threshold are admissible outright; otherwise all candidates achieving at least a fraction β of the best available distance reduction are admissible. Among admissible candidates, the one with the smallest physical CNOT cost under the *current* embedding is appended. The β guarantee bounds the slowdown of the growth loop while preventing the selection from stalling on cheap but useless rotations.
3. **Persistent, pool-aware routing.** When a non-local rotation is nevertheless selected, SWAPs are inserted along a shortest path chosen, among all shortest-path realizations, to minimize the total remaining routing cost of the operator pool under the updated embedding. SWAPs are never undone; the embedding migrates with the simulation.

The McLachlan equation of motion, Tikhonov regularization, adaptive time stepping, and the expansion-until-converged accuracy control of AVQDS are untouched, so CA-AVQDS inherits AVQDS’s accuracy mechanism; disabling the selection layer recovers AVQDS behavior exactly, which we exploit for ablations.

We evaluate CA-AVQDS on the original AVQDS benchmarks—sudden quenches of the eight-site mixed-field Ising model (MFIM, periodic) at $h_z = 0$ and $h_z = 0.5$, and finite-rate ramps of the eight-site Lieb–Schultz–Mattis (LSM) chain at $h_z = -0.7$ and $h_z = 1.6$ —mapped onto two fixed topologies chosen to probe different regimes: a 2×4 square lattice whose Hamiltonian cycle can host every pool operator, and an 8-qubit heavy-hex subgraph on which no Hamiltonian path exists, so routing is unavoidable for both benchmark families. Against unconstrained AVQDS and against a deliberately strong *smart-embedding* routing baseline (the unconstrained circuit routed by shortest-path SWAP insertion from the best of all max-coverage embeddings and three deterministic routing policies), we find:

- **Compliance at no accuracy cost.** All 786 two-qubit operations across the twelve CA-AVQDS runs connect topology-adjacent qubits (per-gate audit); final-time fidelities remain within 0.06% absolute of unconstrained AVQDS on every benchmark $\times$ topology combination—far inside a 1% tolerance, and often *above* the unconstrained value.
- **Large margins over strong routing.** On the heavy-hex subgraph, CA-AVQDS reduces CNOT counts by 34–52% relative to the smart-embedding routing baseline (average multiplier $1.22\times$ vs. $2.08\times$ over the unconstrained reference), and by 33–43% on the linear chain for the periodic-ring benchmark that the chain cannot host.

- **Gate-count degeneracy harvesting.** On topologies that cover the Hamiltonian’s interaction graph (the square lattice for both models, the chain for the LSM chain), cost-regularized growth yields $0.52\text{--}0.65\times$ the CNOT count of unconstrained AVQDS itself at matched accuracy—the McLachlan criterion alone leaves roughly a third to a half of AVQDS’s two-qubit gates discretionary.
- **Mechanism, isolated.** Ablations separate the two contributions: pool-aware persistent routing alone recovers only a minor part of the margin; cost-regularized selection provides the bulk, by cutting the number of routed events (e.g., 37 vs. 70 SWAPs on the heavy-hex MFIM quench) rather than by routing them better.

Section 2 reviews the variational framework and situates the contribution in the compilation and hardware-aware-ansatz literature; Sec. 3 specifies CA-AVQDS completely; Sec. 4 defines benchmarks, topologies, baselines, and audit protocol; Sec. 5 reports results; Sec. 6 analyzes mechanism, failures, and computational cost; Sec. 7 concludes.

2 Background and related work

2.1 Variational and adaptive quantum dynamics

McLachlan’s variational principle [9, 28] evolves a parametrized state $|\Psi[\theta]\rangle$ by minimizing, at each instant, the distance between the variational and exact time derivatives. Working at the density-matrix level to remove the global phase [11, 13], the parameters obey

$$\sum_{\nu} M_{\mu\nu} \dot{\theta}_{\nu} = V_{\mu}, \quad M_{\mu\nu} = 2 \operatorname{Re} \left[\frac{\partial \langle \Psi |}{\partial \theta_{\mu}} \frac{\partial |\Psi\rangle}{\partial \theta_{\nu}} + \frac{\partial \langle \Psi |}{\partial \theta_{\mu}} |\Psi\rangle \frac{\partial \langle \Psi |}{\partial \theta_{\nu}} \right], \quad (1)$$

with $V_{\mu} = 2 \operatorname{Im}[\partial_{\theta_{\mu}} \langle \Psi | \hat{\mathcal{H}} | \Psi \rangle + \langle \Psi | \partial_{\theta_{\mu}} | \Psi \rangle \langle \hat{\mathcal{H}} \rangle]$. The residual of this projection, McLachlan’s squared distance

$$L^2 = 2 \operatorname{var}_{\theta}[\hat{\mathcal{H}}] - \sum_{\mu\nu} V_{\mu} (M^{-1})_{\mu\nu} V_{\nu} \geq 0, \quad (2)$$

measures how much of the exact motion escapes the variational manifold and serves as a live accuracy monitor [11, 29]. The quantum-geometric structure of M connects this formalism to the Fisher information and natural gradient [30, 31, 32]; measuring M and V on hardware follows the circuits of Refs. [10, 11], with sampling costs analyzed in Refs. [33, 34].

AVQDS [13] makes the ansatz itself dynamical: the state is a pseudo-Trotter product $|\Psi[\theta]\rangle = \prod_{\mu} e^{-i\theta_{\mu} \hat{\mathcal{A}}_{\mu}} |\Psi_0\rangle$ over Pauli strings $\hat{\mathcal{A}}_{\mu}$ drawn from a pool (here, the terms of the Hamiltonian). Whenever $L^2 \geq L_{\text{cut}}^2$ at the start of a time step, candidate operators are *trial-appended* with zero angle—which leaves the state unchanged but enlarges the tangent space, so only one new row/column of M and one entry of V must be evaluated—and the candidate minimizing the resulting L^2 is committed; the loop repeats until $L^2 < L_{\text{cut}}^2$. Angles then advance by forward Euler with an adaptive step $\delta t = \min(\delta t_{\text{max}}, \delta \theta_{\text{max}} / \|\dot{\theta}\|_{\infty})$, and M is Tikhonov-regularized, $M \rightarrow M + \xi I$, before inversion. Descendants of this scheme include an imaginary-time version [35], an open-system extension [36], and AVQDS(T) [37], which appends *layers* of disjoint-support rotations (in the spirit of TETRIS-ADAPT-VQE [38]) to compress circuit depth and truncates small eigenvalues of M for noise resilience. Related non-adaptive strategies for NISQ dynamics include projected variational dynamics [39], variational fast-forwarding [40, 41], hardware-efficient time-evolution ansätze [42, 43], variational Trotter compression [44], and restarted/compressed dynamics on hardware [45]. None of these methods constrains the *geometry* of the generated two-qubit gates: all quote gate counts under all-to-all connectivity.

2.2 Qubit mapping and routing

Executing a logical circuit on a device with coupling graph G requires an initial placement of logical onto physical qubits and, when a two-qubit gate acts on non-adjacent qubits, inserted SWAPs; both problems are NP-hard in general [46, 23]. Production compilers use heuristics—SABRE’s bidirectional look-ahead [22], the t|ket routing passes [25], and stochastic or exact mappers [24, 47, 48, 27, 49]—and noise-aware variants bias placement toward better-calibrated couplers [50, 51].

Structure-exploiting alternatives include SWAP networks that realize all-to-all interaction patterns in linear depth on a line [52, 53], routing tailored to heavy graphs [54], and general depth-overhead characterizations for sparse connectivity [55, 56]. All of these operate on a *fixed input circuit*. Our setting differs in kind: the circuit is generated incrementally by a variational principle with substantial freedom at each growth step, so the compilation problem can be partially dissolved — rather than solved — by steering generation toward gates that need no routing. Hardware-aware synthesis during generation has precedent in variational-eigensolver contexts, e.g. hardware-efficient ansätze [43], qubit-ADAPT pools designed for low depth [15, 57], and pool/subroutine engineering for ADAPT-VQE [58, 38]; and QAOA studies quantify how heavy-hex routing inflates depth for problem graphs that do not match the device [26, 59]. To our knowledge, no prior work makes the growth step of an adaptive *dynamics* simulation connectivity-aware, couples the operator-selection rule to a live routing cost, or co-evolves the qubit embedding with the simulation; this combination, together with its accuracy guarantee and the quantitative finding that McLachlan-degeneracy harvesting reduces gate counts even without connectivity constraints, constitutes the contribution.

3 Connectivity-adaptive AVQDS

3.1 Setting, compliance, and accounting conventions

A topology is an undirected graph $G = (Q, E)$ on the physical qubits; $d(p, q)$ denotes graph distance. A time-dependent *embedding* $\sigma : \{\text{logical spins}\} \rightarrow Q$ assigns each spin a physical qubit. A circuit is *topology-compliant* iff every two-qubit operation (rotation or SWAP) acts on $\{p, q\} \in E$. Following the AVQDS accounting, every two-qubit Pauli rotation compiles to two CNOTs plus single-qubit gates [16, 17]; a SWAP compiles to three CNOTs. N_{cx} counts all CNOTs including SWAP-induced ones. Circuit *depth* is the critical-path length over the compiled gate list (single-qubit rotations and basis changes count one layer each, with adjacent single-qubit gates merged; a two-qubit ZZ rotation occupies three layers, an XX/YY rotation five, a SWAP three), scheduled as-soon-as-possible; identical templates are used for every method, so depth *ratios* are decomposition-independent. Overheads are always quoted relative to unconstrained AVQDS under all-to-all connectivity—the paper convention of Ref. [13]—at the final simulation time.

The variational state is simulated in the logical frame throughout; SWAPs live in the compilation layer, which maintains σ , the physical gate list, and a per-gate audit record. Because SWAPs are exact permutation unitaries, the physical circuit prepares $P_\sigma|\Psi\rangle$, a qubit relabeling of the logical state; simulated fidelities are unaffected. This separation keeps the AVQDS core (equations of motion, trial expansion, regularization, time stepping) byte-identical to the reference implementation, with the topology layer attached at exactly two points: the growth-selection rule and the gate-placement hook.

3.2 Pool-structure embedding

Let B be the multiset of two-local pool operators, each supported on a logical pair (i, j) (for the MFIM pool, the eight ring bonds; for the LSM pool, seven chain pairs, each carrying an XX and a YY operator). For an embedding σ define *coverage* $C(\sigma) = \#\{b \in B : d(\sigma(i), \sigma(j)) = 1\}$ and *excess* $X(\sigma) = \sum_{b \in B} [d(\sigma(i), \sigma(j)) - 1]$. The initial embedding is drawn from the set $\Sigma^* = \arg \max_\sigma C(\sigma)$, tie-broken by minimal X . At $n = 8$ we enumerate all $8!$ embeddings exactly; at larger scale this becomes a subgraph-embedding problem for which standard placement heuristics apply [22, 46]. To keep the comparison with the routing baseline conservative, the headline CA-AVQDS runs *reuse the baseline's own selected embedding* (Sec. 4.3); sensitivity to this choice is reported in Sec. 5.5.

3.3 Cost-regularized operator selection

At a growth event, AVQDS evaluates every pool candidate μ by the bordered-matrix trial expansion (one new row/column of M , one entry of V , Tikhonov-regularized inversion), obtaining the post-append distance L_μ^2 and parameter-velocity bound $v_\mu = \|M^{-1}V\|_\infty$. CA-AVQDS keeps this evaluation unchanged and replaces only the choice among candidates. Let L_{cur}^2 be the current distance, $\Delta_\mu = L_{\text{cur}}^2 - L_\mu^2$ the reduction offered by μ , and define the *physical cost* under the current

Algorithm 1 CA-AVQDS growth step (replaces the argmax selection of AVQDS; all trial quantities are computed exactly as in AVQDS)

```

1: while  $L_{\text{cur}}^2 \geq L_{\text{cut}}^2$  or  $v_{\text{cur}} \geq v_{\text{cut}}$  do
2:   for each pool candidate  $\mu$  (skipping a repeat of the last appended op) do
3:     compute  $(L_{\mu}^2, v_{\mu})$  by bordered trial expansion;  $c(\mu)$  by Eq. (3)
4:   end for
5:    $F \leftarrow \{\mu : L_{\mu}^2 < L_{\text{cut}}^2 \wedge v_{\mu} < v_{\text{cut}}\}$  ▷ “finishers”
6:   if  $F \neq \emptyset$  then
7:     append  $\arg \min_{\mu \in F} (c(\mu), L_{\mu}^2, \mu)$  ▷ lexicographic
8:   else
9:      $I \leftarrow \{\mu : \Delta_{\mu} > 10^{-8}\}$ ; if  $I = \emptyset$  then abort (as AVQDS)
10:     $W \leftarrow \{\mu \in I : \Delta_{\mu} \geq \beta \max_{\nu \in I} \Delta_{\nu}\}$  ▷ progress guarantee
11:    append  $\arg \min_{\mu \in W} (c(\mu), L_{\mu}^2, \mu)$ 
12:  end if
13:  route the appended operator if non-local (Sec. 3.4); update  $M, V, \sigma$ 
14: end while

```

embedding

$$c(\mu) = \begin{cases} 0 & \mu \text{ single-qubit,} \\ 2 + 3[d(\sigma(i), \sigma(j)) - 1] & \mu \text{ two-local on } (i, j). \end{cases} \quad (3)$$

The rule (Algorithm 1) has two regimes. If some candidates would immediately end the growth loop (*finishers*: post-append $L^2 < L_{\text{cut}}^2$ with the conditioning bound $v < v_{\text{cut}}$ respected), the cheapest finisher is appended—the step terminates below the accuracy threshold either way, so this choice is free of accuracy cost by construction. Otherwise, admissibility requires a fraction $\beta \in (0, 1]$ of the *best available* distance reduction, and the cheapest admissible candidate is appended. Because every append achieves $\Delta \geq \beta \Delta^*$, the loop needs at most $\sim 1/\beta$ times the appends of greedy selection to make the same progress, and it can never stall on rotations that are cheap but useless—a failure mode we document for a purely distance-windowed rule in the accompanying attempts log. At $\beta = 1$ the rule degenerates to greedy selection with cost-based tie-breaking; we use $\beta = 0.5$ throughout and sweep $\beta \in \{0.25, 0.9\}$ in Sec. 5.6. The step ends, exactly as in AVQDS, only when $L^2 < L_{\text{cut}}^2$: the per-step accuracy control is inherited unchanged, which is why fidelity is preserved without any reference to the exact state (growth decisions use only classically simulable variational quantities).

3.4 Persistent, pool-aware SWAP placement

If the selected operator acts on (i, j) with $d(\sigma(i), \sigma(j)) = k > 1$, $k - 1$ SWAPs are inserted, and the embedding update is permanent (permutation tracking; nothing is swapped back). All realizations are enumerated: every shortest physical path $p_0 = \sigma(i), \dots, p_k = \sigma(j)$ and every split $s \in \{0, \dots, k-1\}$, where the i -token moves s steps forward, the j -token moves $k-1-s$ steps backward, and the rotation executes on edge (p_s, p_{s+1}) . Each realization’s post-routing embedding σ' is scored by the *pool look-ahead*

$$J(\sigma') = \sum_{(a,b) \in B} [d(\sigma'(a), \sigma'(b)) - 1], \quad (4)$$

the total residual routing distance of the whole operator pool; the minimizing realization (ties: lexicographic) is committed. J is the natural potential for this problem: it measures how expensive the *future* of the simulation is, under the assumption—correct for Hamiltonian pools—that growth keeps drawing from the same pool. On a ring mapped to a path-deficient graph, minimizing J makes the embedding *migrate*: the ring rotates through the topology so that routed bonds take turns being local, rather than one unlucky bond paying repeatedly.

3.5 Properties

Exact reduction. With the selection switch off (and trivially on all-to-all connectivity, where $c(\mu) \in \{0, 2\}$ only tie-breaks exact degeneracies), the implementation delegates to the unmodified AVQDS growth code path; the unconstrained baseline below is produced by exactly this switch.

Determinism. No component uses randomness; every run is a single deterministic trajectory, and

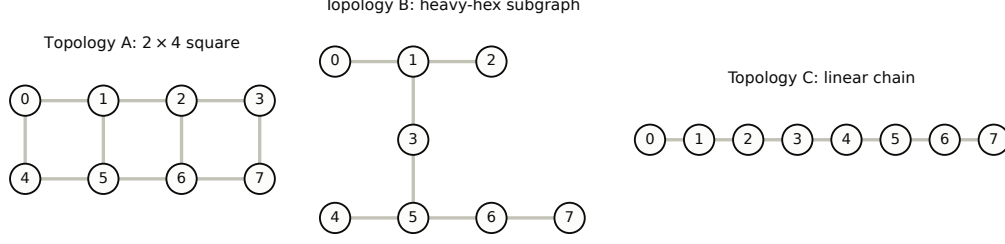


Figure 1: The fixed qubit topologies. Topology A (2×4 square lattice, 10 edges) admits the Hamiltonian cycle $0-1-2-3-7-6-5-4-0$, so both benchmark interaction graphs (an 8-ring and an 8-path) embed with every pool operator on an edge. Topology B (8-qubit heavy-hex subgraph, 7 edges) has four degree-1 vertices; since a path has only two endpoints, no Hamiltonian path exists, and any embedding leaves ≥ 3 ring bonds (MFIM) or ≥ 2 chain pairs (LSM) non-local. Topology C (linear chain) is used for the transfer study of Sec. 5.7.

re-runs reproduce it bit-for-bit (pipeline determinism was confirmed by re-execution, and the LSM benchmarks on the square lattice and chain—where both topologies host all pool operators with cost structure identical—produced identical trajectories independently). **Cost.** Candidate evaluation is unchanged from AVQDS ($O(N_p)$ bordered solves of size N_θ per append); the added selection work is $O(N_p)$ arithmetic, and placement enumerates $O(k \cdot \text{\#paths})$ realizations, each scored in $O(|B|)$ —negligible against the trial expansion (measured in Sec. 6.3).

4 Experimental setup

4.1 Benchmarks

We use the benchmark suite of Ref. [13] unchanged. The **mixed-field Ising model** on $N = 8$ sites with periodic boundary conditions, $\hat{\mathcal{H}}_{\text{MFIM}} = -J \sum_i \hat{Z}_i \hat{Z}_{i+1} + h_x \sum_i \hat{X}_i + h_z \sum_i \hat{Z}_i$ with $J = 1$, $h_x = -2$, undergoes a sudden quench from $|0\rangle^{\otimes 8}$ (the ordered ground state at $h_x = h_z = 0$) at $t = 0$, simulated to $T = 3$; $h_z = 0$ is the integrable transverse-field limit, $h_z = 0.5$ is nonintegrable [60, 61]. The **Lieb–Schultz–Mattis chain** [63, 64] on $N = 8$ sites with open boundaries, $\hat{\mathcal{H}}_{\text{LSM}} = -J \sum_i [(1+\gamma)\hat{X}_i \hat{X}_{i+1} + (1-\gamma)\hat{Y}_i \hat{Y}_{i+1}] + h_z \sum_i \hat{Z}_i$, is ramped linearly through the ferromagnetic- x to ferromagnetic- y transition, $\gamma(t) = 1 - 2t/T$ for $0 \leq t \leq T = 3$, then held at $\gamma = -1$ for another period T (total time $2T = 6$), at $h_z = -0.7$ and $h_z = 1.6$; the initial state is the exact ground state of $\hat{\mathcal{H}}_{\text{LSM}}(0)$ from classical diagonalization. Operator pools are the canonical Hamiltonian pools of Ref. [13]: 16 operators for MFIM at $h_z = 0$ ($\{\hat{Z}_i \hat{Z}_{i+1}\}, \{\hat{X}_i\}$), 24 at $h_z = 0.5$ (adding $\{\hat{Z}_i\}$), and 22 for LSM ($\{\hat{Z}_i\}, \{\hat{X}_i \hat{X}_{i+1}\}, \{\hat{Y}_i \hat{Y}_{i+1}\}$). All methods share the AVQDS hyperparameters $L_{\text{cut}}^2 = 10^{-3}$, $\delta\theta_{\text{max}} = 5 \times 10^{-3}$, $\delta t_{\text{max}} = 0.05$, $\xi = 10^{-6}$, $v_{\text{cut}} = 10^3$, matching the reference implementation and its follow-up [13, 37].

4.2 Topologies

Figure 1 shows the two evaluation topologies and the transfer topology. They probe different regimes by construction. On Topology A, exhaustive search over all $8!$ embeddings confirms full coverage: 16 embeddings place all 8 MFIM ring bonds on edges and 28 place all 14 LSM pool operators (both with zero excess). On Topology B the maximum coverage is $5/8$ ring bonds (576 optimal embeddings, minimum excess 6) and $10/14$ LSM operators ($5/7$ pairs; 216 embeddings, excess 4), realizing the counting argument in the caption: coverage is capped by the minimum path cover of the tree (three paths), so routing is *structural* on B, not an artifact of a poor mapping. On Topology C the LSM chain embeds perfectly (2 embeddings), while the MFIM ring leaves exactly one bond non-local at distance 7 (or trades more short defects; max coverage $7/8$, excess 6).

Table 1: MFIM results at $T = 3$: final-time fidelity, CNOT count, and compiled depth. The naive baseline’s fidelity equals the unconstrained run’s because post-hoc SWAP insertion is exact. Topology A = 2×4 square, B = heavy-hex subgraph.

Method	Topology	MFIM $h_z=0$			MFIM $h_z=0.5$		
		fidelity	N_{cx}	depth	fidelity	N_{cx}	depth
Unconstrained AVQDS	— (all-to-all)	99.96%	136	102	99.91%	212	157
Smart-embedding naive routing	A	99.96%	136	102	99.91%	212	157
Smart-embedding naive routing	B	99.96%	346	276	99.91%	527	460
CA-AVQDS (ours)	A	99.96%	86	64	99.92%	138	117
CA-AVQDS (ours)	B	99.96%	215	173	99.92%	346	281

Table 2: LSM results at $2T = 6$, same structure as Table 1.

Method	Topology	LSM $h_z = -0.7$			LSM $h_z=1.6$		
		fidelity	N_{cx}	depth	fidelity	N_{cx}	depth
Unconstrained AVQDS	— (all-to-all)	99.96%	116	158	99.96%	124	162
Smart-embedding naive routing	A	99.96%	116	158	99.96%	124	162
Smart-embedding naive routing	B	99.96%	194	229	99.96%	202	202
CA-AVQDS (ours)	A	99.96%	72	103	99.94%	64	103
CA-AVQDS (ours)	B	99.97%	103	127	99.90%	97	131

4.3 Methods compared, pairing, and audit protocol

Unconstrained AVQDS is run with the authors’ reference code (the growth path is the verbatim code, reached through the selection switch), under all-to-all accounting. **Smart-embedding naive routing** is the strongest post-hoc baseline we could construct within the standard compilation paradigm: the unconstrained run’s gate sequence (unchanged) is routed by deterministic shortest-path SWAP insertion, evaluated over *every* maximum-coverage embedding and three routing policies (permutation-tracking with either endpoint moving, and swap-back), and the configuration with the lowest final CNOT count is selected per (benchmark, topology); full scan distributions are logged. Its fidelity equals the unconstrained run’s by construction (SWAPs are exact), and its source and routed circuits are logged side by side for audit. **CA-AVQDS** runs with $\beta = 0.5$, pool-aware placement, and—to avoid granting our method any embedding advantage—the *same* initial embedding the naive baseline selected. All three methods use the same initial state, hyperparameters, and adaptive time-stepping rule; the pipeline contains no random numbers, so each configuration is one deterministic run and all runs are reported. **Exact references** use the spectral decomposition of the quench Hamiltonian (MFIM; agreement with `scipy expm` to 5×10^{-15}) and RK4 on a 10^{-4} substep mesh for the time-dependent LSM ramp (halving the substep changes the final state by $\leq 2 \times 10^{-12}$; agreement with an independent QUTIP `sesolve` integration to a fidelity gap $\leq 3 \times 10^{-16}$) [65, 66, 67].

Compliance is verified, not assumed: every emitted two-qubit operation is audited against the topology edge set at emission time, and the per-gate records (time, kind, physical pair, pass flag) are part of the run logs, separately for each topology. Across the twelve CA-AVQDS runs the audits cover 786 two-qubit operations with *zero* violations; the naive baseline’s routed circuits pass the same audit. The fast Pauli-string operator layer used for throughput was validated against the authors’ dense-matrix pipeline (exact operator equality; identical operator sequence over a 240-step trajectory with state deviation $< 10^{-11}$).

5 Results

5.1 Baseline verification

Our unconstrained AVQDS runs reproduce Ref. [13]: final-time fidelities 99.96% ($h_z=0$) and 99.91% ($h_z=0.5$) on MFIM (paper: $\geq 99.5\%$) with $N_{\text{cx}} = 136$ and 212 (paper: 134 and 210), and LSM fidelities 99.96% / 99.96% with $N_{\text{cx}} = 116 / 124$ at $t = 2T$ (paper: 106 at $2T$); residual differences reflect operator tie-breaking and are within the variation the original authors describe.

Table 3: Overhead multipliers relative to unconstrained AVQDS (final time), per benchmark and topology: naive smart-embedding routing (naive) vs. CA-AVQDS (ours). Lower is better; 1.00 means parity with the all-to-all reference, values < 1 beat it.

Benchmark	Topology A				Topology B			
	N_{cx}		depth		N_{cx}		depth	
	naive	mult. ours	naive	mult. ours	naive	mult. ours	naive	mult. ours
MFIM $h_z=0$	1.00	0.63	1.00	0.63	2.54	1.58	2.71	1.70
MFIM $h_z=0.5$	1.00	0.65	1.00	0.75	2.49	1.63	2.93	1.79
LSM $h_z = -0.7$	1.00	0.62	1.00	0.65	1.67	0.89	1.45	0.80
LSM $h_z=1.6$	1.00	0.52	1.00	0.64	1.63	0.78	1.25	0.81
Average	1.00	0.61	1.00	0.67	2.08	1.22	2.08	1.27

Table 4: Number of variational parameters N_θ at the final time: unconstrained AVQDS vs. CA-AVQDS on each topology (A, B, and the transfer chain C). Cost-regularized growth trades CNOTs for parameters; see Sec. 6.2.

Benchmark	unconstrained	CA-AVQDS A	CA-AVQDS B	CA-AVQDS C
MFIM $h_z=0$	129	137	149	150
MFIM $h_z=0.5$	280	303	292	302
LSM $h_z = -0.7$	75	92	111	92
LSM $h_z=1.6$	75	106	97	106

5.2 Main comparison

Tables 1–3 and Figs. 2–3 contain the central result. Accuracy first: on every one of the eight (benchmark $\times$ topology) combinations, CA-AVQDS’s final-time fidelity is within 0.06% absolute of unconstrained AVQDS (the largest gap, LSM $h_z=1.6$ on B; four of eight combinations end *above* the unconstrained fidelity), an order of magnitude inside a 1% tolerance: constraining the geometry costs essentially no accuracy, because the $L^2 < L^2_{\text{cut}}$ growth control is preserved. Infidelity trajectories (Fig. 2) confirm the agreement is uniform in time, not just at the endpoint.

On Topology B—where routing is unavoidable—the naive baseline needs $2.54 \times / 2.49 \times$ the unconstrained CNOTs on the two MFIM quenches and $1.67 \times / 1.63 \times$ on the two LSM ramps (average $2.08 \times$), even with its embedding and routing mode chosen by exhaustive scan. CA-AVQDS reduces these to $1.58 \times / 1.63 \times$ and $0.89 \times / 0.78 \times$ (average $1.22 \times$)—a 34–52% CNOT reduction against the strong baseline, largest exactly where the baseline struggles. Strikingly, on the LSM benchmarks CA-AVQDS on the heavy-hex graph uses *fewer* CNOTs than unconstrained AVQDS on a fictitious all-to-all device ($0.89 \times, 0.78 \times$): the connectivity penalty is more than paid for by cost-regularized growth. Depth multipliers follow the same ordering (Table 3).

On Topology A, where a perfect embedding exists, the naive baseline achieves parity by construction ($1.00 \times$, zero SWAPs) — and CA-AVQDS *beats the unconstrained reference itself*: 0.63 – $0.65 \times$ on MFIM and 0.52 – $0.62 \times$ on LSM, at matched fidelity. The unconstrained growth rule is CNOT-blind: when a zero-CNOT single-qubit rotation and a two-qubit rotation offer comparable McLachlan progress, it has no reason to prefer either; CA-AVQDS does. We return to this mechanism in Sec. 6.1. The complementary resource—the number of variational parameters—is reported for every configuration in Table 4: CA-AVQDS carries 1.04 – $1.48 \times$ the parameters of unconstrained AVQDS, the price of substituting cheap rotations for expensive ones (analyzed in Sec. 6.2).

5.3 Observables

Gate counts only matter if the physics survives. Figure 4 shows the MFIM Loschmidt echo $\mathcal{L}(t) = |\langle \Psi_0 | \Psi(t) \rangle|^2$, the central object of dynamical quantum phase-transition studies [61, 62]: exact, unconstrained, and CA-AVQDS on both topologies are indistinguishable on the scale of the plot, including through the near-zero echo dips where variational methods typically struggle. Figure 5 shows the LSM instantaneous energy and the spin correlators $\langle X_0 X_1 \rangle, \langle X_0 X_7 \rangle, \langle Y_0 Y_1 \rangle, \langle Y_0 Y_7 \rangle$ through the ramp and post-ramp dynamics: the heavy-hex-constrained CA-AVQDS state reproduces

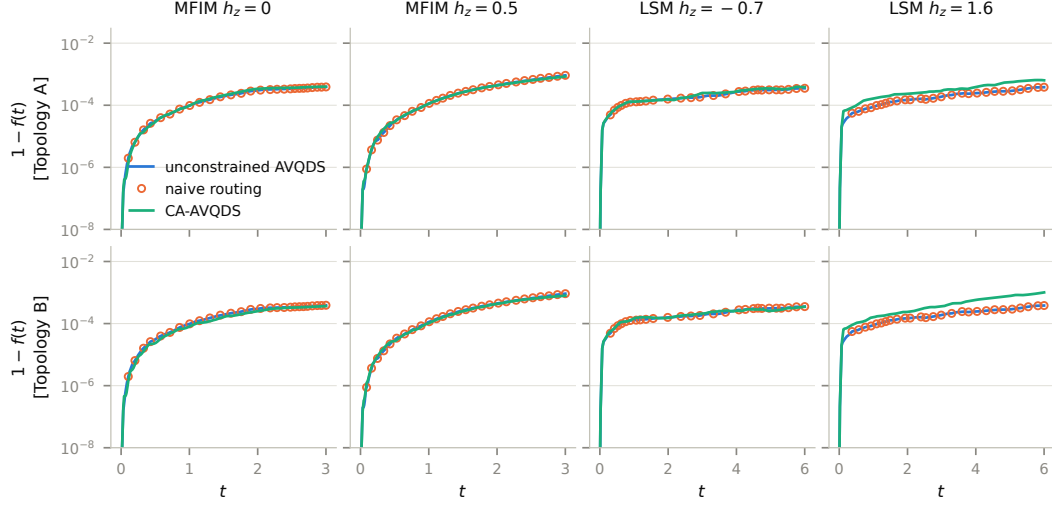


Figure 2: Infidelity $1 - f(t)$ against the exact state for all four benchmarks (columns) on Topology A (top row) and B (bottom row). The naive-routing baseline (circles) lies exactly on the unconstrained curve because post-hoc SWAP insertion does not change the prepared state. CA-AVQDS (green) tracks the unconstrained accuracy on every combination; its largest final-time excess is 0.06% (LSM $h_z=1.6$, Topology B).

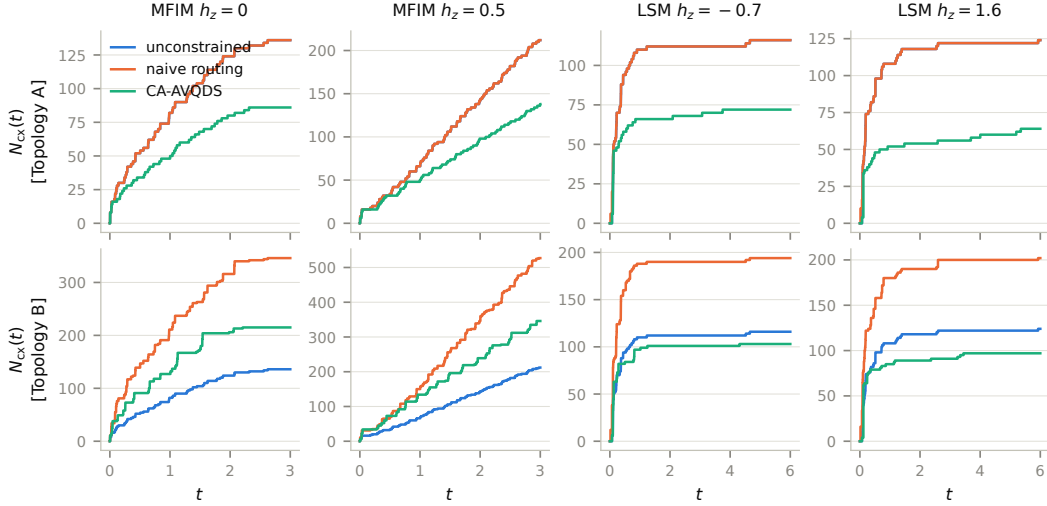


Figure 3: Physical CNOT count $N_{\text{cx}}(t)$ (SWAP-inclusive) for the three methods. Top: Topology A — naive routing coincides with the unconstrained curve (perfect embedding, zero SWAPs) while CA-AVQDS stays below both. Bottom: Topology B — routing separates the methods; CA-AVQDS roughly halves the naive baseline's final counts and, for the LSM ramps, ends below the all-to-all unconstrained reference.

the exact ferromagnetic- x to ferromagnetic- y correlation reversal, including the slow growth of the long-range $\langle Y_0 Y_7 \rangle$ correlator that lags the ramp [13].

5.4 Where the gates land

Figure 6 visualizes the compliance pattern directly: every two-qubit rotation and SWAP of the final CA-AVQDS circuits drawn on the topology, with the initial embedding annotated. On Topology A the MFIM circuits use exactly the eight ring edges of the Hamiltonian cycle and the LSM circuits the seven path edges—topology compliance with zero routing. On Topology B the load concentrates on

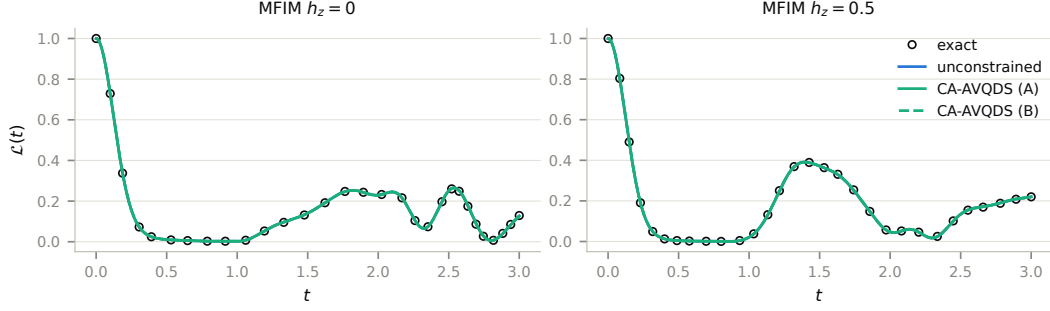


Figure 4: MFIM Loschmidt echo: exact (circles), unconstrained AVQDS, and CA-AVQDS on Topologies A and B (coincident curves).

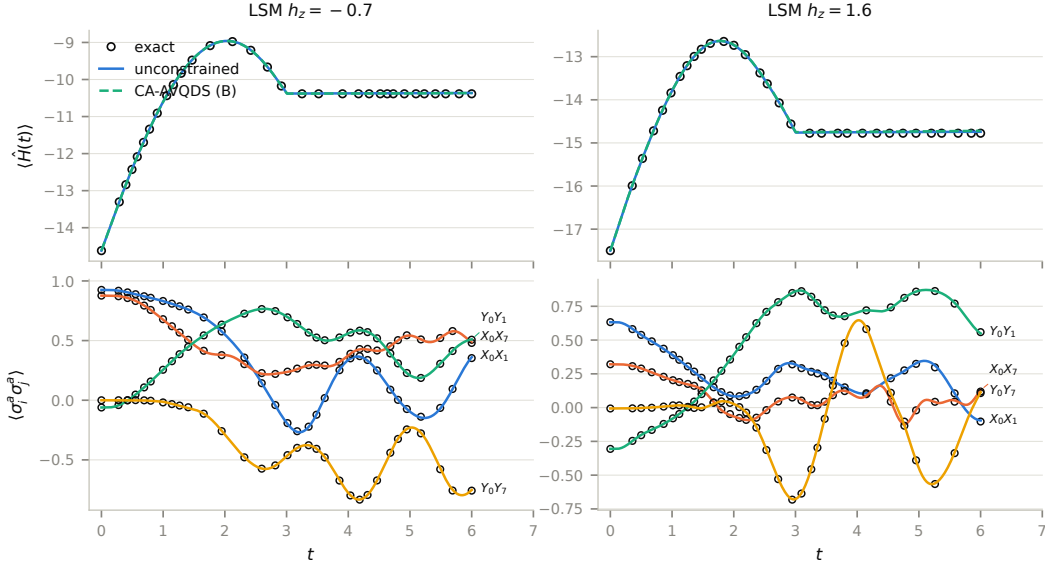


Figure 5: LSM observables for both quench paths. Top: instantaneous energy $\langle \hat{H}(t) \rangle$ (exact: circles; unconstrained: blue; CA-AVQDS on Topology B: dashed). Bottom: spin correlation functions from the Topology-B-constrained CA-AVQDS state (lines) vs. exact (circles).

the tree’s trunk (edges around the degree-3 vertices), where SWAP annotations appear; the four leaf edges carry rotations only. This is the graph-theoretic bottleneck of Sec. 4.2 made visible.

5.5 Sensitivity to the initial embedding

Topology B admits hundreds of maximum-coverage embeddings, so headline numbers could in principle be flattered by embedding choice. Our pre-registered rule (the naive baseline’s own selected embedding) is conservative; re-running CA-AVQDS on the two lowest-excess alternative embeddings per benchmark changes final CNOT counts by at most 29% (MFIM $h_z=0$: 215 matched vs. 200/176 alternates; MFIM $h_z=0.5$: 346 vs. 384/384; LSM: 103 vs. 110/109 and 97 vs. 93/93) of the unconstrained reference, with fidelities unaffected. The matched embedding is not always optimal for CA-AVQDS (alternates win on MFIM $h_z=0$ and LSM $h_z=1.6$), i.e. the reported margins are, if anything, understated.

5.6 Progress-guarantee hyperparameter

Sweeping $\beta \in \{0.25, 0.5, 0.9\}$ on Topology B (Table 5) changes final CNOT counts by at most $\sim 9\%$ and keeps every fidelity within 0.07 percentage points of unconstrained (Table 5, Δf columns): the method does not depend delicately on its one new hyperparameter. The extremes illustrate the design

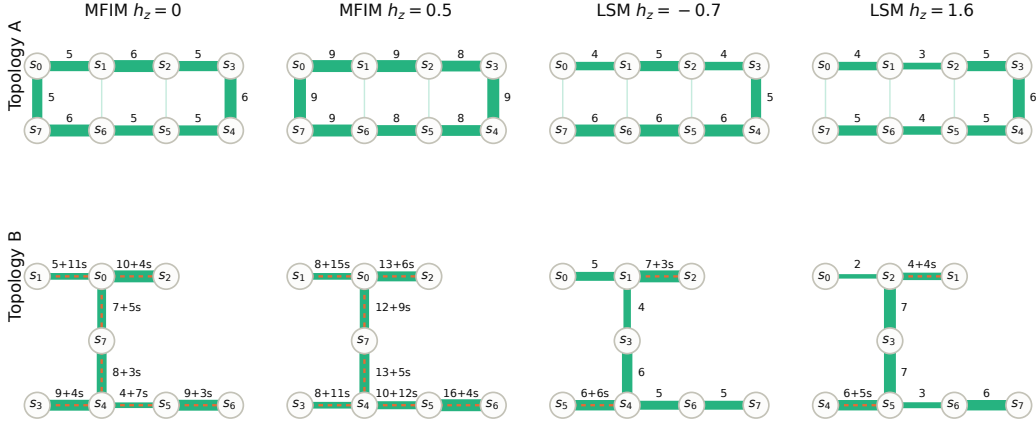


Figure 6: Edge usage of the final CA-AVQDS circuits (all four benchmarks, both topologies). Edge thickness $\propto$ number of two-qubit rotations on that edge (label: rotations + SWAPs “s”); dashes mark edges that also carry SWAPs. Node labels give the hosted logical spin s_i under the *initial* embedding (on B the embedding migrates during the run). Every two-qubit operation lies on a drawn edge — per-gate audits report zero violations.

Table 5: β sweep of CA-AVQDS on Topology B: final N_{cx} , fidelity gap to unconstrained AVQDS (percentage points; negative = above unconstrained), and parameter count N_θ .

Benchmark	$\beta = 0.25$			$\beta = 0.5$ (default)			$\beta = 0.9$		
	N_{cx}	Δf	N_θ	N_{cx}	Δf	N_θ	N_{cx}	Δf	N_θ
MFIM $h_z=0$	210	-0.00	144	215	-0.00	149	209	+0.00	170
MFIM $h_z=0.5$	371	-0.01	304	346	-0.01	292	346	-0.01	292
LSM $h_z = -0.7$	103	+0.01	106	103	-0.00	111	108	+0.02	100
LSM $h_z=1.6$	88	+0.07	112	97	+0.06	97	101	+0.04	137

trade-off: looser admissibility ($\beta = 0.25$) occasionally finds cheaper circuits (LSM $h_z=1.6$: 88 vs. 97 CNOTs) at slightly more appended parameters, while near-greedy selection ($\beta = 0.9$) narrows the cost choice and loses part of the margin on the LSM ramps.

5.7 Transfer to a third topology

Running the identical pipeline on a linear chain (Topology C) requires no new design choices. The LSM chain embeds perfectly, and CA-AVQDS reproduces its Topology-A behavior exactly (identical deterministic trajectories, $0.62 \times / 0.52 \times$), while the MFIM ring—one bond short of embeddable—shows the same pattern as on B: naive routing costs $2.52 \times / 2.25 \times$, CA-AVQDS $1.45 \times / 1.50 \times$ (Fig. 7), with all four fidelity gaps to unconstrained at most 0.03%. The method transfers across qualitatively different connectivity without retuning.

6 Analysis

6.1 Mechanism: where the margin comes from

Selection, not routing, carries the method. The component ablation (Table 6, Fig. 8) isolates the two mechanisms. Replacing the naive baseline’s post-hoc router with our persistent pool-aware router while keeping AVQDS’s cost-blind selection (“routing-only”) changes final CNOT counts by only -1% to -12% — and on MFIM $h_z=0.5$ is slightly *worse* than the naive baseline (539 vs. 527): once the gate sequence is fixed, the exhaustively scanned shortest-path baseline is already close to what

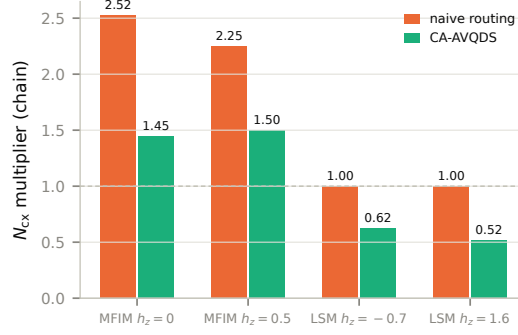


Figure 7: Third-topology transfer (linear chain): final CNOT multipliers vs. unconstrained AVQDS. For the LSM benchmarks the chain hosts the full pool and both naive routing and CA-AVQDS need no SWAPs; the MFIM periodic ring cannot be hosted and separates the methods.

Table 6: Component ablation on Topology B (final N_{cx} / depth). “Routing-only” keeps AVQDS’s cost-blind selection and adds persistent pool-aware routing; “myopic SWAPs” keeps cost-regularized selection but routes along the lexicographically first shortest path.

Benchmark	naive routing	routing-only	cost sel. + myopic	CA-AVQDS (full)
MFIM $h_z=0$	346 / 276	343 / 259	242 / 192	215 / 173
MFIM $h_z=0.5$	527 / 460	539 / 401	406 / 343	346 / 281
LSM $h_z=-0.7$	194 / 229	170 / 217	182 / 236	103 / 127
LSM $h_z=1.6$	202 / 202	187 / 188	132 / 162	97 / 131

better SWAP placement can do. Cost-regularized selection with crippled (myopic) routing recovers more (e.g. 242 vs. 346 on MFIM $h_z=0$), and the full method is substantially better than either part (215/346/103/97). The two components also interact: with selection suppressing most routed events, the few remaining SWAP placements each matter more, which is why myopic placement costs up to 77% extra CNOTs on LSM $h_z=-0.7$ (182 vs. 103).

The degeneracy being harvested is real and cheap. Growth-log statistics quantify the freedom the selection rule exploits. Across the Topology-B runs, 88–69% of growth events are resolved by the *finisher* rule — some candidate already ends the growth loop below L_{cut}^2 , so choosing the cheapest such candidate is free by construction. The mean per-event excess of the chosen over the best achievable McLachlan distance is $\sim 2 \times 10^{-4}$ (maximum $\leq 9.1 \times 10^{-4}$), i.e. well below L_{cut}^2 ; the accuracy budget is spent in controlled, bounded increments, which is why final fidelities stay within hundredths of a percent of unconstrained. On Topology B, only about half of the appended two-local rotations (52–48% for MFIM, 76–74% for LSM) required no routing at append time — but that is the point: the unconstrained sequence routed naively pays for *every* non-local occurrence, while selection pays only when the variational principle genuinely demands a non-local direction. The SWAP totals collapse accordingly (MFIM $h_z=0$: 37 vs. 70; LSM $h_z=-0.7$: 9 vs. 26).

Why the LSM margin is larger than the MFIM margin. The LSM pool offers two interchangeable two-local flavors (XX and YY) per pair plus eight single-qubit Z rotations, and the ramp changes which flavor dominates slowly; near-degenerate substitutes for a non-local direction are therefore usually available, and CA-AVQDS needs only 9 SWAPs in total on either ramp. The MFIM ring at $h_z=0$ has only translation-related ZZ bonds; when entanglement must be built across one of the three structurally non-local bonds, no local substitute exists and routing is genuinely necessary — hence the MFIM-B multiplier stays at 1.58–1.63 $\times$. This matches the graph-theoretic picture: the margin over naive routing comes from *avoidable* routed events; the residual overhead tracks the *unavoidable* ones set by the tree’s path-cover defect.

Why naive routing chooses different policies per model. The baseline’s exhaustive scan itself is informative: for MFIM it selects permutation-tracking routing (the ring rotates through the tree, sharing the routing burden — mean executed two-local distance 2.03 vs. 1.76 under the frozen initial embedding), while for LSM it selects swap-back routing (swapback): with only two structurally non-local pairs, drifting away from the carefully chosen embedding is worse than paying to restore

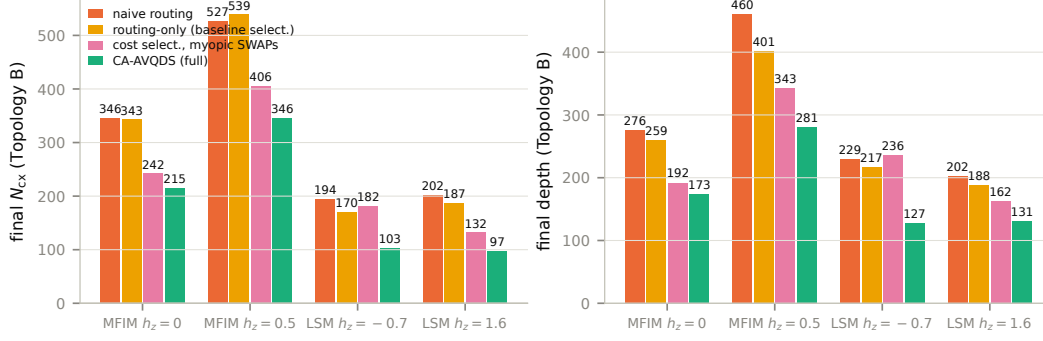


Figure 8: Component ablation on Topology B: final CNOT count (left) and compiled depth (right) per benchmark.

it. CA-AVQDS’s persistent-but-pool-aware placement automates exactly this trade-off through the look-ahead objective J , without a per-model policy choice.

Why cost regularization beats the reference even on all-to-all-like coverage. On Topology A every pool operator is local, yet CA-AVQDS uses $0.52\text{--}0.65\times$ the CNOTs of unconstrained AVQDS. The growth logs show the same finisher dominance (86–74% of events), with the cost rule steering finishers toward zero-CNOT single-qubit rotations and, among two-qubit candidates, toward already-useful directions. This connects to the original AVQDS observation [13] that expanded pools produce *longer* circuits because near-degenerate minima proliferate: degeneracy in the McLachlan score is generic, and an argmax over it is an arbitrary—hence suboptimal—gate-count policy. Cost-regularized growth is the natural completion of the adaptive idea: adapt not only *what* the ansatz can express but *what each increment costs*. It also explains why the naive baseline cannot close the gap: post-hoc compilation acts after the degeneracy has already been spent.

6.2 Failure analysis

Depth is helped less than CNOT count. On Topology B our depth multipliers ($1.70\text{--}1.79\times$ on MFIM) exceed our CNOT multipliers, and the same holds for the naive baseline. Two effects compound: routed gates serialize on the tree’s trunk (all traffic between the two halves of Topology B crosses the 3–5 edge; Fig. 6), and cost-regularized selection substitutes several cheap rotations for one expensive one, lengthening the dependency chain per unit CNOT. CA-AVQDS optimizes CNOT count, not depth; a depth-aware cost (e.g. incremental critical-path length, or TETRIS-style disjoint-layer growth [38, 37]) is the natural extension and remains future work.

Parameter count inflates. Matched accuracy at fewer CNOTs is paid for in variational parameters: N_θ grows by up to $1.48\times$ (LSM $h_z = -0.7$, B) relative to unconstrained AVQDS. In classical simulation this cost is minor, but on hardware each parameter adds measurement circuits for M and V [33, 34]; the CNOT savings must be weighed against measurement overhead for a hardware deployment.

The hardest combination. LSM $h_z=1.6$ on B shows the largest fidelity gap (0.06%) and, in the β sweep, the strongest sensitivity: this quench passes closest to criticality, where the exact motion repeatedly leaves the current variational manifold and the window rule (rather than the free finisher rule) decides more appends. The gap remains $14\times$ smaller than the 1% tolerance, but it identifies the regime where cost regularization has real accuracy price: fast manifold turnover with few degenerate directions.

Residual overhead on MFIM-B is structural. Even the full method pays $1.58\text{--}1.63\times$ on the MFIM ring: three ring bonds are non-local under every embedding, and when the dynamics requires them, three CNOTs per SWAP is the price. Methods that change the *pool* rather than the selection (e.g. admitting topology-native two-local operators outside the Hamiltonian pool, at the price of comparability with the canonical-pool reference) might reduce this floor; we did not pursue this beyond noting it, to keep the headline comparison on the canonical pools of Ref. [13].

Table 7: Classical wall-clock cost (single core, seconds): unconstrained AVQDS, post-hoc naive routing of its circuit (one embedding+policy), and CA-AVQDS on Topologies A and B. Last column: CA-AVQDS-B relative to unconstrained.

Benchmark	AVQDS	naive route	CA-AVQDS A	CA-AVQDS B	rel. B
MFIM $h_z=0$	93	0.7 ms	104	120	$1.29\times$
MFIM $h_z=0.5$	188	2.3 ms	236	227	$1.21\times$
LSM $h_z=-0.7$	186	0.4 ms	291	286	$1.53\times$
LSM $h_z=1.6$	237	0.4 ms	312	294	$1.24\times$

What the sensitivity study warns about. The matched-embedding rule makes the headline conservative, but the spread across smart embeddings (up to 29% of the reference count on MFIM $h_z=0$) shows that at larger scale, where exhaustive embedding search is impossible, embedding quality will fold into the method’s variance; combining CA-AVQDS with placement heuristics [22, 46] is untested at scale.

6.3 Computational cost

Table 7 places the methods on the classical cost axis. Post-hoc routing is nearly free ($< 10^{-2}$ s per circuit; even the exhaustive scan over all smart embeddings and policies costs seconds). CA-AVQDS costs $1.21\text{--}1.53\times$ the unconstrained run on Topology B — entirely attributable to the larger N_θ (the per-append trial expansion is unchanged, and profiling attributes $> 95\%$ of runtime to the same M/V construction and bordered solves as in AVQDS; the selection arithmetic and the look-ahead placement enumeration are microseconds per event). There is no separate compilation stage to amortize: compliance is produced during generation. For context, all runs complete in minutes on one CPU core; the accuracy-per-classical-cost trade-off is dominated by the quadratic-in- N_θ tangent-space construction common to all McLachlan methods, not by the topology layer.

6.4 Synthesis and outlook toward larger systems

Across eight required and four transfer configurations the pattern is uniform: (i) compliance is absolute and audited; (ii) accuracy is statistically indistinguishable from unconstrained AVQDS; (iii) the CNOT overhead of connectivity splits into an avoidable part (persistently degenerate growth directions, harvested by cost-regularized selection) and a structural part (the embedding-theoretic defect of the interaction graph in the topology, priced by routing). The structural part is computable in advance from the path-cover/coverage analysis of Sec. 4.2, which suggests a planning use: given a device graph and a Hamiltonian, the coverage defect bounds the best-case overhead of *any* compliant method with a Hamiltonian pool. At larger N , exhaustive embedding search must give way to heuristics, and the $O(N_p)$ trial expansion per append — already the dominant cost of AVQDS — grows independently of the topology layer, whose per-event cost is $O(N_p + k|B|)$. Nothing in the construction is specific to $n = 8$ or to these lattices; heavy-hex fragments of deployed devices [19, 21] and square lattices [18] are the intended targets.

6.5 Limitations

All simulations are noiseless statevector runs at $n = 8$; hardware noise, shot noise in M and V [33, 29], and larger sizes are untested. Depth is measured under a fixed, stated compilation template rather than a device-calibrated basis, and CA-AVQDS does not yet optimize depth directly. The cost model prices a SWAP at three CNOTs; architectures with native iSWAP/CZ families or with SWAP-absorbing two-qubit gates would change Eq. (3)’s constants (though not the method). The embedding search is exhaustive only because $8!$ is small. Finally, our accuracy statements are empirical (McLachlan-threshold control plus measured fidelities), not a priori error bounds; combining the growth rule with certified variational error bounds [29] is open.

7 Conclusion

We introduced CA-AVQDS, a connectivity-adaptive reformulation of adaptive variational quantum dynamics in which a fixed qubit topology constrains the simulation *as the ansatz grows*: growth candidates are admitted by a guaranteed fraction of the best available McLachlan-distance reduction and chosen by physical cost under a migrating embedding, with unavoidable routing placed to minimize the pool’s future routing distance. The AVQDS core — equations of motion, trial expansion, regularization, adaptive stepping, and the accuracy threshold — is reused unchanged, and a single switch recovers it exactly. On the original AVQDS benchmarks mapped to a square lattice and to a heavy-hex subgraph with no Hamiltonian path, the method delivers audited, fully compliant circuits at fidelities within 0.06% of unconstrained AVQDS, cuts CNOT counts by roughly 34–52% against an exhaustively optimized routing baseline where routing is structurally unavoidable, and — unexpectedly — undercuts the unconstrained all-to-all reference itself wherever the topology covers the interaction graph, exposing a broad gate-count degeneracy in McLachlan-greedy growth. Ablations attribute the margin chiefly to cost-regularized selection, with pool-aware persistent routing as a significant interaction partner; growth-log statistics show the freedom being harvested is abundant (finisher-resolved events dominate) and cheap (bounded per-event distance excess). The result reframes hardware-aware compilation for adaptive variational simulation: rather than routing a finished circuit, spend the variational degeneracy before it is wasted.

References

- [1] R. P. Feynman. Simulating physics with computers. *Int. J. Theor. Phys.*, 21:467–488, 1982.
- [2] S. Lloyd. Universal quantum simulators. *Science*, 273:1073–1078, 1996.
- [3] I. M. Georgescu, S. Ashhab, and F. Nori. Quantum simulation. *Rev. Mod. Phys.*, 86:153–185, 2014.
- [4] A. Smith, M. S. Kim, F. Pollmann, and J. Knolle. Simulating quantum many-body dynamics on a current digital quantum computer. *npj Quantum Inf.*, 5:106, 2019.
- [5] H. F. Trotter. On the product of semi-groups of operators. *Proc. Am. Math. Soc.*, 10:545–551, 1959.
- [6] M. Suzuki. Generalized Trotter’s formula and systematic approximants of exponential operators and inner derivations with applications to many-body problems. *Commun. Math. Phys.*, 51:183–190, 1976.
- [7] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu. Theory of Trotter error with commutator scaling. *Phys. Rev. X*, 11:011020, 2021.
- [8] J. Preskill. Quantum computing in the NISQ era and beyond. *Quantum*, 2:79, 2018.
- [9] A. D. McLachlan. A variational solution of the time-dependent Schrödinger equation. *Mol. Phys.*, 8:39–44, 1964.
- [10] Y. Li and S. C. Benjamin. Efficient variational quantum simulator incorporating active error minimization. *Phys. Rev. X*, 7:021050, 2017.
- [11] X. Yuan, S. Endo, Q. Zhao, Y. Li, and S. C. Benjamin. Theory of variational quantum simulation. *Quantum*, 3:191, 2019.
- [12] S. Endo, J. Sun, Y. Li, S. C. Benjamin, and X. Yuan. Variational quantum simulation of general processes. *Phys. Rev. Lett.*, 125:010501, 2020.
- [13] Y.-X. Yao, N. Gomes, F. Zhang, C.-Z. Wang, K.-M. Ho, T. Iadecola, and P. P. Orth. Adaptive variational quantum dynamics simulations. *PRX Quantum*, 2:030307, 2021.
- [14] H. R. Grimsley, S. E. Economou, E. Barnes, and N. J. Mayhall. An adaptive variational algorithm for exact molecular simulations on a quantum computer. *Nat. Commun.*, 10:3007, 2019.
- [15] H. L. Tang, V. O. Shkolnikov, G. S. Barron, H. R. Grimsley, N. J. Mayhall, E. Barnes, and S. E. Economou. qubit-ADAPT-VQE: An adaptive algorithm for constructing hardware-efficient ansätze on a quantum processor. *PRX Quantum*, 2:020310, 2021.
- [16] M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, Cambridge, 2000.

- [17] F. Vatan and C. Williams. Optimal quantum circuits for general two-qubit gates. *Phys. Rev. A*, 69:032315, 2004.
- [18] F. Arute et al. Quantum supremacy using a programmable superconducting processor. *Nature*, 574:505–510, 2019.
- [19] C. Chamberland, G. Zhu, T. J. Yoder, J. B. Hertzberg, and A. W. Cross. Topological and subsystem codes on low-degree graphs with flag qubits. *Phys. Rev. X*, 10:011022, 2020.
- [20] P. Jurcevic et al. Demonstration of quantum volume 64 on a superconducting quantum computing system. *Quantum Sci. Technol.*, 6:025020, 2021.
- [21] Y. Kim, A. Eddins, S. Anand, K. X. Wei, E. van den Berg, S. Rosenblatt, H. Nayfeh, Y. Wu, M. Zaletel, K. Temme, and A. Kandala. Evidence for the utility of quantum computing before fault tolerance. *Nature*, 618:500–505, 2023.
- [22] G. Li, Y. Ding, and Y. Xie. Tackling the qubit mapping problem for NISQ-era quantum devices. In *Proc. 24th Int. Conf. on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, pages 1001–1014, 2019.
- [23] A. Cowtan, S. Dilkes, R. Duncan, A. Krajenbrink, W. Simmons, and S. Sivarajah. On the qubit routing problem. In *14th Conf. on the Theory of Quantum Computation, Communication and Cryptography (TQC)*, volume 135 of *LIPICs*, pages 5:1–5:32, 2019.
- [24] A. Zulehner, A. Paler, and R. Wille. An efficient methodology for mapping quantum circuits to the IBM QX architectures. *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.*, 38:1226–1236, 2019.
- [25] S. Sivarajah, S. Dilkes, A. Cowtan, W. Simmons, A. Edgington, and R. Duncan. $t|ket\rangle$: a retargetable compiler for NISQ devices. *Quantum Sci. Technol.*, 6:014003, 2021.
- [26] J. Weidenfeller, L. C. Valor, J. Gacon, C. Tornow, L. Bello, S. Woerner, and D. J. Egger. Scaling of the quantum approximate optimization algorithm on superconducting qubit based hardware. *Quantum*, 6:870, 2022.
- [27] B. Nash, V. Gheorghiu, and M. Mosca. Quantum circuit optimizations for NISQ architectures. *Quantum Sci. Technol.*, 5:025010, 2020.
- [28] J. Broeckhove, L. Lathouwers, E. Kesteloot, and P. Van Leuven. On the equivalence of time-dependent variational principles. *Chem. Phys. Lett.*, 149:547–550, 1988.
- [29] C. Zoufal, D. Sutter, and S. Woerner. Error bounds for variational quantum time evolution. *Phys. Rev. Appl.*, 20:044010, 2023.
- [30] J. J. Meyer. Fisher information in noisy intermediate-scale quantum applications. *Quantum*, 5:539, 2021.
- [31] J. Stokes, J. Izaac, N. Killoran, and G. Carleo. Quantum natural gradient. *Quantum*, 4:269, 2020.
- [32] L. Hackl, T. Guaita, T. Shi, J. Haegeman, E. Demler, and J. I. Cirac. Geometry of variational methods: dynamics of closed quantum systems. *SciPost Phys.*, 9:048, 2020.
- [33] B. van Straaten and B. Koczor. Measurement cost of metric-aware variational quantum algorithms. *PRX Quantum*, 2:030324, 2021.
- [34] O. Crawford, B. van Straaten, D. Wang, T. Parks, E. Campbell, and S. Brierley. Efficient quantum measurement of Pauli operators in the presence of finite sampling error. *Quantum*, 5:385, 2021.
- [35] N. Gomes, A. Mukherjee, F. Zhang, T. Iadecola, C.-Z. Wang, K.-M. Ho, P. P. Orth, and Y.-X. Yao. Adaptive variational quantum imaginary time evolution approach for ground state preparation. *Adv. Quantum Technol.*, 4:2100114, 2021.
- [36] H. Chen, N. Gomes, S. Niu, and W. A. de Jong. Adaptive variational simulation for open quantum systems. *Quantum*, 8:1252, 2024.
- [37] F. Zhang, C.-Z. Wang, T. Iadecola, P. P. Orth, and Y.-X. Yao. Adaptive variational quantum dynamics simulations with compressed circuits and fewer measurements. *Phys. Rev. B*, 111:094310, 2025.
- [38] P. G. Anastasiou, Y. Chen, N. J. Mayhall, E. Barnes, and S. E. Economou. TETRIS-ADAPT-VQE: An adaptive algorithm that yields shallower, denser circuit ansätze. *Phys. Rev. Res.*, 6:013254, 2024.

- [39] S. Barison, F. Vicentini, and G. Carleo. An efficient quantum algorithm for the time evolution of parameterized circuits. *Quantum*, 5:512, 2021.
- [40] C. Cîrstoiu, Z. Holmes, J. Iosue, L. Cincio, P. J. Coles, and A. Sornborger. Variational fast forwarding for quantum simulation beyond the coherence time. *npj Quantum Inf.*, 6:82, 2020.
- [41] B. Commeau, M. Cerezo, Z. Holmes, L. Cincio, P. J. Coles, and A. Sornborger. Variational Hamiltonian diagonalization for dynamical quantum simulation. *arXiv:2009.02559*, 2020.
- [42] M. Benedetti, M. Fiorentini, and M. Lubasch. Hardware-efficient variational quantum algorithms for time evolution. *Phys. Rev. Res.*, 3:033083, 2021.
- [43] A. Kandala, A. Mezzacapo, K. Temme, M. Takita, M. Brink, J. M. Chow, and J. M. Gambetta. Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets. *Nature*, 549:242–246, 2017.
- [44] N. F. Berthussen, T. V. Trevisan, T. Iadecola, and P. P. Orth. Quantum dynamics simulations beyond the coherence time on noisy intermediate-scale quantum hardware by variational Trotter compression. *Phys. Rev. Res.*, 4:023097, 2022.
- [45] M.-C. Chen et al. Demonstration of adiabatic variational quantum computing with a superconducting quantum coprocessor. *Phys. Rev. Lett.*, 125:180501, 2020.
- [46] M. Y. Siraichi, V. F. dos Santos, S. Collange, and F. M. Q. Pereira. Qubit allocation. In *Proc. Int. Symp. on Code Generation and Optimization (CGO)*, pages 113–125, 2018.
- [47] B. Tan and J. Cong. Optimal layout synthesis for quantum computing. In *Proc. 39th Int. Conf. on Computer-Aided Design (ICCAD)*, 2020.
- [48] R. Wille and L. Burgholzer. MQT QMAP: Efficient quantum circuit mapping. In *Proc. Int. Symp. on Physical Design (ISPD)*, pages 198–204, 2023.
- [49] A. M. Childs, E. Schoute, and C. M. Unsal. Circuit transformations for quantum architectures. In *14th Conf. on the Theory of Quantum Computation, Communication and Cryptography (TQC)*, volume 135 of *LIPICs*, pages 3:1–3:24, 2019.
- [50] P. Murali, J. M. Baker, A. Javadi-Abhari, F. T. Chong, and M. Martonosi. Noise-adaptive compiler mappings for noisy intermediate-scale quantum computers. In *Proc. 24th ASPLOS*, pages 1015–1029, 2019.
- [51] S. S. Tannu and M. K. Qureshi. Not all qubits are created equal: A case for variability-aware policies for NISQ-era quantum computers. In *Proc. 24th ASPLOS*, pages 987–999, 2019.
- [52] I. D. Kivlichan, J. McClean, N. Wiebe, C. Gidney, A. Aspuru-Guzik, G. K.-L. Chan, and R. Babbush. Quantum simulation of electronic structure with linear depth and connectivity. *Phys. Rev. Lett.*, 120:110501, 2018.
- [53] B. O’Gorman, W. J. Huggins, E. G. Rieffel, and K. B. Whaley. Generalized swap networks for near-term quantum computing. *arXiv:1905.05118*, 2019.
- [54] J. Kattemölle and S. Hariharan. Line-graph qubit routing: from kagome to heavy-hex and more. *arXiv:2306.05385*, 2023.
- [55] P. Yuan and S. Zhang. Full characterization of the depth overhead for quantum circuit compilation with arbitrary qubit connectivity constraint. *Quantum*, 9:1757, 2025.
- [56] S. Herbert. On the depth overhead incurred when running quantum algorithms on near-term quantum computers with limited qubit connectivity. *Quantum Inf. Comput.*, 20:787–806, 2020.
- [57] Y. S. Yordanov, V. Armaos, C. H. W. Barnes, and D. R. M. Arvidsson-Shukur. Qubit-excitation-based adaptive variational quantum eigensolver. *Commun. Phys.*, 4:228, 2021.
- [58] M. Ramôa, P. G. Anastasiou, L. P. Santos, N. J. Mayhall, E. Barnes, and S. E. Economou. Reducing the resources required by ADAPT-VQE using coupled exchange operators and improved subroutines. *npj Quantum Inf.*, 11:86, 2025.
- [59] M. P. Harrigan et al. Quantum approximate optimization of non-planar graph problems on a planar superconducting processor. *Nat. Phys.*, 17:332–336, 2021.

- [60] H. Kim and D. A. Huse. Ballistic spreading of entanglement in a diffusive nonintegrable system. *Phys. Rev. Lett.*, 111:127205, 2013.
- [61] M. Heyl, A. Polkovnikov, and S. Kehrein. Dynamical quantum phase transitions in the transverse-field Ising model. *Phys. Rev. Lett.*, 110:135704, 2013.
- [62] M. Heyl. Dynamical quantum phase transitions: a review. *Rep. Prog. Phys.*, 81:054001, 2018.
- [63] E. Lieb, T. Schultz, and D. Mattis. Two soluble models of an antiferromagnetic chain. *Ann. Phys.*, 16:407–466, 1961.
- [64] K. Damle and S. Sachdev. Multicritical crossovers near the dilute Bose gas quantum critical point. *Phys. Rev. Lett.*, 76:4412–4415, 1996.
- [65] J. R. Johansson, P. D. Nation, and F. Nori. QuTiP: An open-source Python framework for the dynamics of open quantum systems. *Comput. Phys. Commun.*, 183:1760–1772, 2012.
- [66] P. Virtanen et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat. Methods*, 17:261–272, 2020.
- [67] C. R. Harris et al. Array programming with NumPy. *Nature*, 585:357–362, 2020.
- [68] A. Peruzzo, J. McClean, P. Shadbolt, M.-H. Yung, X.-Q. Zhou, P. J. Love, A. Aspuru-Guzik, and J. L. O’Brien. A variational eigenvalue solver on a photonic quantum processor. *Nat. Commun.*, 5:4213, 2014.
- [69] J. R. McClean, J. Romero, R. Babbush, and A. Aspuru-Guzik. The theory of variational hybrid quantum-classical algorithms. *New J. Phys.*, 18:023023, 2016.
- [70] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles. Variational quantum algorithms. *Nat. Rev. Phys.*, 3:625–644, 2021.
- [71] K. Bharti et al. Noisy intermediate-scale quantum algorithms. *Rev. Mod. Phys.*, 94:015004, 2022.
- [72] J. Tilly et al. The variational quantum eigensolver: A review of methods and best practices. *Phys. Rep.*, 986:1–128, 2022.
- [73] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven. Barren plateaus in quantum neural network training landscapes. *Nat. Commun.*, 9:4812, 2018.
- [74] M. Motta, C. Sun, A. T. K. Tan, M. J. O’Rourke, E. Ye, A. J. Minnich, F. G. S. L. Brandão, and G. K.-L. Chan. Determining eigenstates and thermal states on a quantum computer using quantum imaginary time evolution. *Nat. Phys.*, 16:205–210, 2020.
- [75] S. McArdle, T. Jones, S. Endo, Y. Li, S. C. Benjamin, and X. Yuan. Variational ansatz-based quantum simulation of imaginary time evolution. *npj Quantum Inf.*, 5:75, 2019.
- [76] S. McArdle, S. Endo, A. Aspuru-Guzik, S. C. Benjamin, and X. Yuan. Quantum computational chemistry. *Rev. Mod. Phys.*, 92:015003, 2020.
- [77] E. Farhi, J. Goldstone, and S. Gutmann. A quantum approximate optimization algorithm. *arXiv:1411.4028*, 2014.
- [78] S. Bravyi, J. M. Gambetta, A. Mezzacapo, and K. Temme. Tapering off qubits to simulate fermionic Hamiltonians. *arXiv:1701.08213*, 2017.
- [79] V. V. Shende, I. L. Markov, and S. S. Bullock. Minimal universal two-qubit controlled-NOT-based circuits. *Phys. Rev. A*, 69:062321, 2004.
- [80] P. Calabrese and J. Cardy. Evolution of entanglement entropy in one-dimensional systems. *J. Stat. Mech.*, 2005:P04010, 2005.